CSA1708-ARTIFICIAL INTELLIGENCE

LAB EXPERIMENTS

Program 3: write the python program for water jug problem

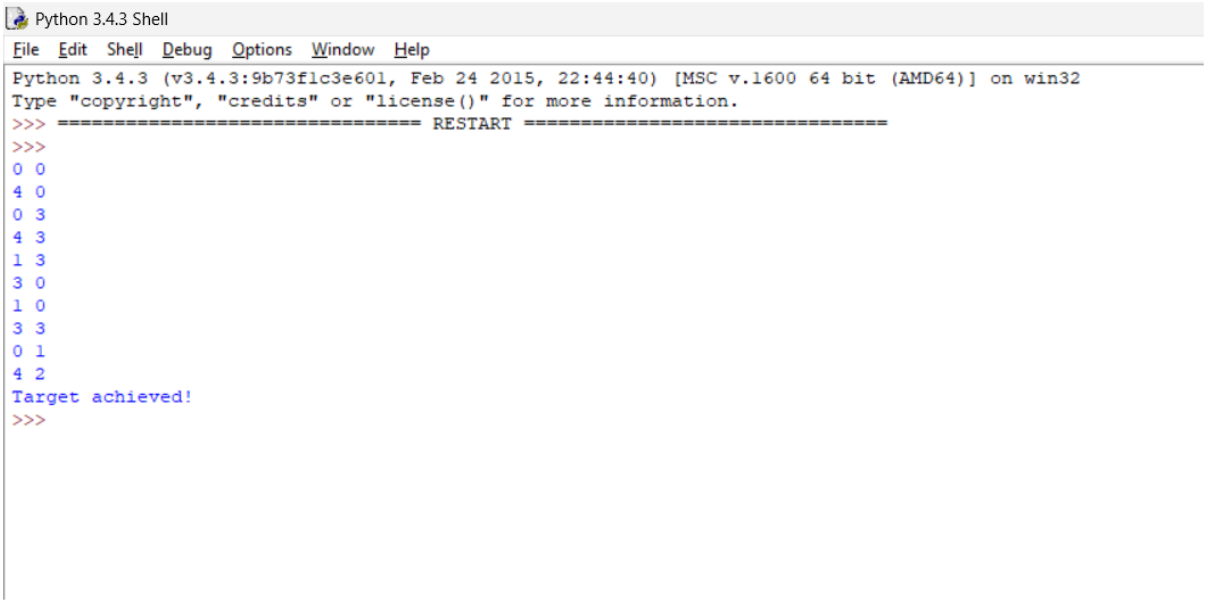
Aim:

To write a Python program to solve the **Water Jug Problem** using the **Breadth-First Search (BFS)** .

Algorithm:

1. Start the program.
2. Define the capacities of two jugs and the target amount.
3. Start with both jugs empty.
4. Generate possible operations: fill, empty, and pour.
5. Store unvisited states in a queue.
6. Repeat until the target amount is obtained.
7. Display the sequence of states.
8. Stop.





Result:

Thus, the **Water Jug Problem was successfully solved using Python**, and the required target amount was obtained.